

Udit Singh Parihar

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📍 Bangalore, India

EDUCATION

International Institute of Information Technology

Hyderabad, India

MS By Research Computer Science; **GPA: 8.67/10.00**

2019 – 2021

Indian Institute of Technology

Jodhpur, India

B. Tech Mechanical Engineering; **GPA: 7.1/10.0**

2014 – 2018

WORK EXPERIENCE

Qualcomm

Bangalore, India

Computer Vision, DL and SLAM Research Engineer

July 2023 – Present

Multi-camera Vision 3DoF Head Tracking (shipped in Google AndroidXR, Samsung Moohan & XREAL Aura):

- Architected multi-camera support in tracker and pose graph optimizer, doubling robustness, RMSE, jitter KPIs; implemented in multi-threaded C++ with tight memory management to hold the real-time budget
- Reformulated the pose graph optimizer to solve only independent variables, halving state and cutting latency 30%
- Root-caused content drift for users in moving vehicles; fixed and merged at Google in 3 weeks, unblocking OTA
- Corrected the covariance propagation formulation, tightening correlation between covariance and true error

Patents (9 Accepted) – Deep Learning & SLAM:

- **Perception under scene change (shipped in AndroidXR):** Depth, feature-consistency and IMU motion-compensation methods for users on moving platforms; reset-decision metric and DL matching under illumination change
- **Deep feature matching:** Hybrid classical-DL tracking with IMU priors; adaptive low-compute matcher with server offload; dual-head network predicting correspondences and uncertainty, fused into EKF pose estimation
- **Scene-aware SLAM:** VLM semantic and user-context priors to adaptively configure SLAM perception; deterministic-power head tracking via multi-map vision-IMU fusion with rotation-only map merging

Visual-Inertial Odometry Optimizations:

- Showed IMU intrinsics and camera-IMU clock offset can be dropped from the EKF-based VIO state vector, saving latency and power, validated over 15 datasets
- Replaced a Hermite spline distortion model with a closed-form warping matrix, cutting the top power hotspot

Deep Feature Matching, Chip Spec & Evaluation Infrastructure:

- Swept 72 sensor and compute configurations to define the minimum viable chip spec for a next-generation XR device (768×768, 132° FOV, 2 Hz, 800 features) at 1° RMSE
- Traced a 50% frame-failure rate at reduced camera FOV to descriptor and matching quality, and showed recovery with DL matchers benchmarked over 14 datasets × 10 methods
- Built a mechanically constrained 4-ChArUco rig for sub-mm ground-truth pose collection where motion capture is unavailable; integrated into CI

OLA Electric

Bangalore, India

Computer Vision, DL and SLAM Research Engineer

July 2021 – July 2023

Autonomous agent development:

- Developed multimodal network to predict Semantic occupancy grid and drivable waypoints directly from camera and GPS Coordinate
- Developed simulation in loop and hardware in loop testing for the autonomous agent
- Converted the pytorch model to TensorRT and developed a ROS wrapper to deploy on Mahindra E2O car achieving final control prediction at 25 HZ, in a zero shot paradigm

Lidar based mapping and localization:

- Extended the Lidar based SLAM LeGO-LOAM for the Ouster lidars and ported ROS1 to ROS2 in C++

- Calibrated the Lidar and IMU/GNSS sensors for extrinsic calibration

Kaggle Image Matching Challenge:

- Won the silver medal in the Kaggle Image Matching Challenge 2022
- Developed an Ensemble of Deep feature matching algorithm of SuperGlue and LoFTR

RESEARCH PUBLICATIONS

1. **Estimation of Appearance and Occupancy Information in Bird's Eye View from Surround Monocular Images**  | *OLA Electric*
International Conference on Robotics and Automation (ICRA), Autonomy 2.0, 2022
Sarthak Sharma, Unnikrishnan R. Nair, Udit Singh Parihar, Midhun Menon S and Srikanth Vidapanakal
2. **RoRD: Rotation-Robust Descriptors and Orthographic Views for Local Feature Matching**  | *IIITH*
International Conference on Intelligent Robots and Systems (IROS), 2021
Udit Singh Parihar, Aniket Gujarathi*, Kinal Mehta*, Satyajit Tourani*, Sourav Garg, Michael Milford and K. Madhava Krishna*
3. **Early Bird: Loop Closures from Opposing Viewpoints for Perceptually-Aliased Indoor Environments**  | *IIITH*
International Conference on Computer Vision Theory and Applications (VISAPP), 2021
Satyajit Tourani, Dhagash Desai*, Udit Singh Parihar*, Sourav Garg, Ravi Kiran Sarvadevabhatla, Michael Milford and K. Madhava Krishna*
4. **Topological Mapping for Manhattan-like Repetitive Environments**  | *IIITH*
International Conference on Robotics and Automation (ICRA), 2020
Sai Shubodh Puligilla, Satyajit Tourani*, Tushar Vaidya*, Udit Singh Parihar*, Ravi Kiran Sarvadevabhatla and K. Madhava Krishna*

PROJECTS

Tutorial on Pose Graph Optimization | [Project Link](#)

Teaching Assistant in Mobile Robotics course | Sep 2020

- Created Open source tutorials for 2D pose graph optimization with loop closure and 3D pose graph optimization with landmarks using g2o library
- Obtained more than 50 stars and forks on GitHub for the tutorials

Development of Robotics Toolbox | [Project Link](#)

Mobile Robotics Coursework | Aug 2019

- Implemented Bundle Adjustment from scratch. Compared performance of Gauss Newton and LM algorithm for optimization
- Implemented Extended Kalman Filter algorithm on the standard "Lost in the Woods" dataset

Development of Parallel Computing Toolbox | [Project Link](#)

Parallel Scientific Computing Coursework | Jan 2019

- Implemented PCA algorithms for image compression using C++/Cuda. Compared performance against MATLAB standard PCA implementation
- Implemented parallel Monte Carlo algorithm for calculation of digits of π using OpenMP and MPI

SKILLS

Programming: C++, Python, C, MATLAB

Libraries: PyTorch, Keras, CUDA, TensorRT, ROS1/ROS2, Ceres, GTSAM

RELEVANT COURSEWORK

Major coursework: Computer Vision, Mobile Robotics, Topics in Applied Optimization, Introduction to Parallel Programming, Deep Learning Theory and Practices, Probability and Statistics, Programming and Data Structures